

Student's Name: _____ Enroll No: _____
Section: _____

Subject:	Artificial Intelligence (LAB)		Date:	25 th March, 2025
Course Code:	CSC-233		Marks:	10
Examination:	Quiz-01	Spring 2025	Semester:	4th
Instructor:	Noreen Khalid		Time:	20 mints

Mapping to CLO	
After successful completion of this course, the students should be able to:	Q#
1. Understand and explain foundational AI principles and algorithms	01, 02
2. Implement and apply AI models to solve computational problems	01, 02
3. Analyze and evaluate AI solutions for efficiency and effectiveness.	0
4. Assess ethical and societal implications of AI technologies.	0

Error handling Questions:

[Marks-03] Mapped with CLO-01 & 02

1. What will happen if you try to predict using a linear regression model before calling .fit()?

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
predictions = model.predict([[1, 2], [3, 4]])
```

- a. Attribute Error
- b. Value Error
- c. Not Fitted Error
- d. No error

2. What error does this NumPy operation cause?

```
import numpy as np
a = np.array([1, 2, 3])
b = np.array([[1, 2], [3, 4]])
result = a + b
```

- a. Shape Error
- b. Value Error
- c. Index Error
- d. No error

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3. What error will the following code produce?

```
import numpy as np
arr = np.array([1, 2, 3])
print(arr[5])
```

- a. Index Error
- b. Type Error
- c. Value Error
- d. No Error

Error Finding Questions:

[Marks-03] Mapped with CLO-01 & 02

- a. The following code intends to concatenate two NumPy arrays, but it results in an error. Identify and fix it:

```
import numpy as np
a = np.array([[1, 2], [3, 4]])
b = np.array([5, 6])
result = np.concatenate((a, b), axis=1)
```

Correction:

- b. The following code attempts to merge two Data Frames, but an error occurs. Find and fix it:

```
1 df1 = pd.DataFrame({'ID': [1, 2, 3], 'Name': ['Alice', 'Bob', 'Charlie']})
2 df2 = pd.DataFrame({'Age': [25, 30, 35], 'City': ['NY', 'LA', 'SF']})
3 merged_df = df1.merge(df2, on='ID')
```

Correction:

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c. Find and fix the error in the following prediction step:

```
from sklearn.linear_model import LinearRegression
import numpy as np

X_train = np.array([[1], [2], [3], [4], [5]])
y_train = np.array([10, 20, 30, 40, 50])

model = LinearRegression()
model.fit(X_train, y_train)

X_test = [6, 7, 8] # What is wrong with this test input?
predictions = model.predict(X_test)
```

Correction:

Output Finding

[Marks-04] Mapped with CLO-01 & 02

- a. Reverse the given NumPy array and print the result. `arr = np.array([1, 2, 3, 4, 5])`
- b. Train a linear regression model on a given dataset and predict the output for $X = 6$.

```
x = np.array([[1], [2], [3], [4], [5]])
y = np.array([2, 4, 6, 8, 10])
```

- c. Train a multiple linear regression model and find the **coefficients** of the model.
- d. Given a NumPy array, find the value at the second-last position.

```
arr = np.array([1, 2, 3, 4, 5])
```

GOODLUCK